

Math 550
Homework 3

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Exercise 1. Let $\mathcal{A} \subset \mathcal{P}(X)$ be an algebra, $\mathcal{A}_\sigma$ the collection of countable unions of sets in $\mathcal{A}$, and $\mathcal{A}_{\sigma\delta}$ the collection of countable intersections of sets in $\mathcal{A}_\sigma$. Let μ_0 be a premeasure on $\mathcal{A}$ and μ^* the induced outer measure.

- (a) For any $E \subset X$ and $\epsilon > 0$ there exists $A \in \mathcal{A}_\sigma$ with $E \subset A$ and $\mu^*(A) \leq \mu^*(E) + \epsilon$.
- (b) If $\mu^*(E) < \infty$, then E is μ^* -measurable if and only if there exists $B \in \mathcal{A}_{\sigma\delta}$ with $E \subset B$ and $\mu^*(B \setminus E) = 0$.
- (c) If μ_0 is σ -finite, the restriction $\mu^*(E) < \infty$ in (b) is superfluous.

Proof. (a) Let $E \subset X$ and $\epsilon > 0$. By the definition of $\mu^*(E)$, we can find $\{A_j\}_{j=1}^\infty \subset \mathcal{A}$ such that $\bigcup_{j=1}^\infty A_j \supset E$ and:

$$\begin{aligned} \mu^*(E) + \epsilon &\geq \sum_{j=1}^\infty \mu_0(A_j) && \text{Property of infimums} \\ &= \sum_{j=1}^\infty \mu^*(A_j) && \text{Because } \mu^*|_{\mathcal{A}} = \mu_0 \\ &\geq \mu^*\left(\bigcup_{j=1}^\infty A_j\right). && \text{Outer measures are subadditive} \end{aligned}$$

(b) $(\Rightarrow)$ Suppose E is μ^* -measurable. By (a), for every $n \in \mathbf{N}$, there exists $A_n \in \mathcal{A}_\sigma$ such that $E \subset A_n$ and $\mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}$. Since $\mu^*(E) < \infty$, we have $\mu^*(A_n) - \mu^*(E) \leq \frac{1}{n}$. Define $B = \bigcap_{n=1}^\infty A_n \in \mathcal{A}_{\sigma\delta}$. Then for all $n \in \mathbf{N}$:

$$\begin{aligned} \mu^*(B \setminus E) &= \mu^*(B \cap E^c) \\ &\leq \mu^*(A_n \cap E^c) && \text{Monotonicity \& } B \subset A_n \ \forall n \\ &= \mu^*(A_n) - \mu^*(A_n \cap E) && \text{Since } E \text{ is } \mu^*\text{-measurable} \\ &= \mu^*(A_n) - \mu^*(E) && \text{Since } E \subset A_n \\ &\leq \frac{1}{n} \end{aligned}$$

Hence $\mu^*(B \setminus E) = 0$. $(\Leftarrow)$ Now let $B \in \mathcal{A}_{\sigma\delta}$ such that $E \subset B$ and $\mu^*(B \setminus E) = 0$. Apply part (a) to the set $B \setminus E$. For each $n \in \mathbf{N}$, there exists $A_n \in \mathcal{A}_\sigma$ such that $B \setminus E \subset A_n$ and $\mu^*(A_n) \leq \mu^*(B \setminus E) + \frac{1}{n} = \frac{1}{n}$. Since $\mathcal{A}_\sigma \subset \mathfrak{M}(\mathcal{A}) \subset \mathcal{M}^*$, and since $\mathcal{M}^*$ is a σ -algebra, the set $\bigcap_{n=1}^\infty A_n$ is measurable. Since $B \setminus E \subset A_n$ for every $n \in \mathbf{N}$, we have $B \setminus E \subset \bigcap_{n=1}^\infty A_n$. Since $\bigcap_{n=1}^\infty A_n \subset A_n$, for every $n \in \mathbf{N}$ we have $\mu^*\left(\bigcap_{n=1}^\infty A_n\right) \leq \mu^*(A_n) \leq \frac{1}{n}$. Hence $\mu^*\left(\bigcap_{n=1}^\infty A_n\right) = 0$. Since $(X, \mathcal{M}^*, \mu^*|_{\mathcal{M}^*})$ is a complete measure space, and since $\bigcap_{n=1}^\infty A_n$ is a null-set, every subset of $\bigcap_{n=1}^\infty A_n$ is measurable. In particular, $B \setminus E$ is measurable. Moreover, $B \in \mathcal{A}_{\sigma\delta} \subset \mathcal{M}^*$, hence B is also measurable. We can write $E = B \setminus (B \setminus E) = B \cap (B \setminus E)^c$. Note

that $B \in \mathcal{M}^*$, and $(B \setminus E)^c \in \mathcal{M}^*$ since $B \setminus E \in \mathcal{M}^*$ and since $\mathcal{M}^*$ is closed under intersections (it is a σ -algebra), hence $E \in \mathcal{M}^*$ meaning that it is measurable.

(c) We did not use the restriction $\mu^*(E) < \infty$ in the $(\Leftarrow)$ direction, hence we only need to prove $(\Rightarrow)$ for when μ_0 is σ -finite. Let $X = \bigcup_{j=1}^{\infty} X_j$. Then $\mu_0(X_i) < \infty$. Define $E_n = E \cap \left(\bigcup_{j=1}^n X_j\right)$. Then $\bigcup_{n=1}^{\infty} E_n = E$.

For every $n \in \mathbf{N}$, apply part (a) to E_n .

There exists $C_n \in \mathcal{A}_\delta$ such that $E_n \subset C_n$ and $\mu^*(C_n) \leq \mu^*(E_n) + \frac{1}{n}$. Define $C = \bigcap_{n=1}^{\infty} C_n$. Then $\mu^*(C) \leq \mu^*(C_n) \leq \mu^*(E_n) + \frac{1}{n}$. Rearranging gives $\mu^*(C) - \mu^*(E_n) \leq \frac{1}{n}$. But since $\mu^*(E_n) \leq \mu^*(E)$, we have $\mu^*(C) - \mu^*(E) \leq \mu^*(C) - \mu^*(E_n) \leq \frac{1}{n}$. Since this holds true for all $n \in \mathbf{N}$, we have $\mu^*(C \setminus E) = 0$. $\square$

Exercise 2. Let μ^* be an outer measure on X induced by the finite premeasure μ_0 . If $E \subset X$, define the *inner measure* of E to be $\mu_*(E) = \mu_0(X) - \mu^*(E^c)$. Show E is μ^* -measurable if and only if $\mu^*(E) = \mu_*(E)$.

Proof. $(\Rightarrow)$ Suppose E is μ^* -measurable. Then E^c is also μ^* -measurable since $\mathcal{M}^*$ is a σ -algebra. We can write:

$$\mu^*(X) = \mu^*(E) + \mu^*(E^c).$$

Since $X \in \mathcal{A}$ and $\mu_0 = \mu^*|_{\mathcal{A}}$, we have:

$$\mu_0(X) = \mu^*(E) + \mu^*(E^c).$$

Subtracting both sides by $\mu^*(E^c)$ gives $\mu_*(E) = \mu^*(E)$.

$(\Leftarrow)$ Now suppose $\mu^*(E) = \mu_0(X) - \mu^*(E^c)$. Again, since $X \in \mathcal{A}$ and $\mu_0 = \mu^*|_{\mathcal{A}}$, we have:

$$\mu^*(E) + \mu^*(E^c) = \mu^*(X) = \mu_0(X) < \infty$$

For each $n \in \mathbf{N}$, Apply Exercise 1, part (a) to E . Then there exists $A_n \in \mathcal{A}_\sigma$ such that $E \subset A_n$ and $\mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}$. Define $B := \bigcap_{n=1}^{\infty} A_n$. Note that B is measurable since $B \in \mathcal{A}_{\sigma\delta} \subset \mathcal{M}^*$. Moreover since $E \subset A_n$ for all $n \in \mathbf{N}$, we have $E \subset B$. Together:

$$\mu^*(E) \leq \mu^*(B) \leq \mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}.$$

Since this holds true for all $n \in \mathbf{N}$, it must be that $\mu^*(E) = \mu^*(B)$. Quickly note:

$$\begin{aligned} \mu^*(B) + \mu^*(B^c) &= \mu^*(X) \\ &= \mu^*(E) + \mu^*(E^c). \end{aligned}$$

Since $\mu^*(B) = \mu^*(E) < \infty$, we have $\mu^*(E^c) = \mu^*(B^c)$. Now by the measurability of B :

$$\begin{aligned} \mu^*(E) + \mu^*(E^c) &= \mu^*(E) + \mu^*(E^c \cap B) + \mu^*(E^c \cap B^c) \\ &= \mu^*(E) + \mu^*(E^c \cap B) + \mu^*(B^c) \\ &= \mu^*(E) + \mu^*(E^c \cap B) + \mu^*(E^c) \end{aligned}$$

Hence $\mu^*(E^c \cap B) = \mu^*(B \setminus E) = 0$. By Exercise 1, part (b), E is μ^* -measurable. $\square$